

Find and Fix: Input, Output, and Function Notation

Directions

- In $f(a) = b$ the number a is the **input** and b is the **output**. Decide which one a question is asking for before you start.
- Several problems show work that a student got wrong. Correct it *and* say what went wrong — both are part of the answer.
- Write the substitution you used, not just the final number.

1. $f(x) = 5x + 1$. Find $f(4)$.

Answer: _____

2. Devin was asked to evaluate $h(x) = 4x - 9$ when the input is -2 . He wrote:

$$h(-2) = 4(-9) - 2 = -38$$

- (a) Work out the correct value of $h(-2)$: _____

- (b) Describe what Devin did wrong.

3. $f(x) = 2x + 7$. Find the value of x for which $f(x) = 15$.

Answer: _____

4. The table gives the function t .

n	1	2	3	4	5
$t(n)$	3	8	13	18	23

$t(4) =$ _____ and the input n for which $t(n) = 8$ is $n =$ _____

5. The table gives the function c .

x	1	2	3	4
$c(x)$	9	14	19	24

Rosa was asked which input gives an output of 19. She answered 19.

(a) Give the correct answer: $x =$ _____

(b) Describe the mistake Rosa made.

6. $f(x) = 3x + 4$ and $g(x) = 4x + 3$. A student claims that $f(2)$ and $g(2)$ must be equal “because the numbers are just swapped”.

$$f(2) = \underline{\hspace{2cm}} \quad g(2) = \underline{\hspace{2cm}} \quad f(2) \underline{\hspace{2cm}} g(2)$$

Explain whether the student’s claim is right, and why.

7. $p(x) = -2x + 11$.

(a) Find $p(5)$: $\underline{\hspace{2cm}}$

(b) Find the value of x for which $p(x) = 5$: $\underline{\hspace{2cm}}$

8. A function d is given by the rule $d(x) = 3x + 2$. Kofi was told that $d(6) = 20$ and was asked what it means. He wrote: “It means 6 is the output when you put in 20.”

(a) Rewrite Kofi’s sentence so that it is correct.

(b) Use the rule to work out $d(6)$ and check that the statement $d(6) = 20$ is true: $\underline{\hspace{2cm}}$